Course Announcements

Due Sunday:

- Q4
- A3
- Project Proposal - we grade from GitHub

Project Proposal

Full project guidelines:
https://github.com/COGS108/Projects/blob/fa26/FinalProject_Guidelines.md#project-proposal

Data Science Questions

Shannon E. Ellis, Ph.D
UC San Diego

Department of Cognitive Science
sellis@ucsd.edu



Formulating Data Science Questions

When you and your group sit down to figure out what you're going to do for your final project in this class, you'll have to formulate a strong question - one that is specific, can be answered with data, and makes clear what exactly is being measured.

1 START HERE

Ask an interesting question

- What is the scientific **goal**?
- What would you do if you had **all the data**?
- What do you want to **predict** or **estimate**?

2

Get the data

- How were the data **sampled**?
- Which data are **relevant**?
- Are there **privacy** issues?

3

Explore the data

- **Plot** the data.
- Are there **anomalies**?
- Are there **patterns**?

4

Model the data

- **Build** a model.
- **Fit** the model.
- **Validate** the model.

5

Communicate the results

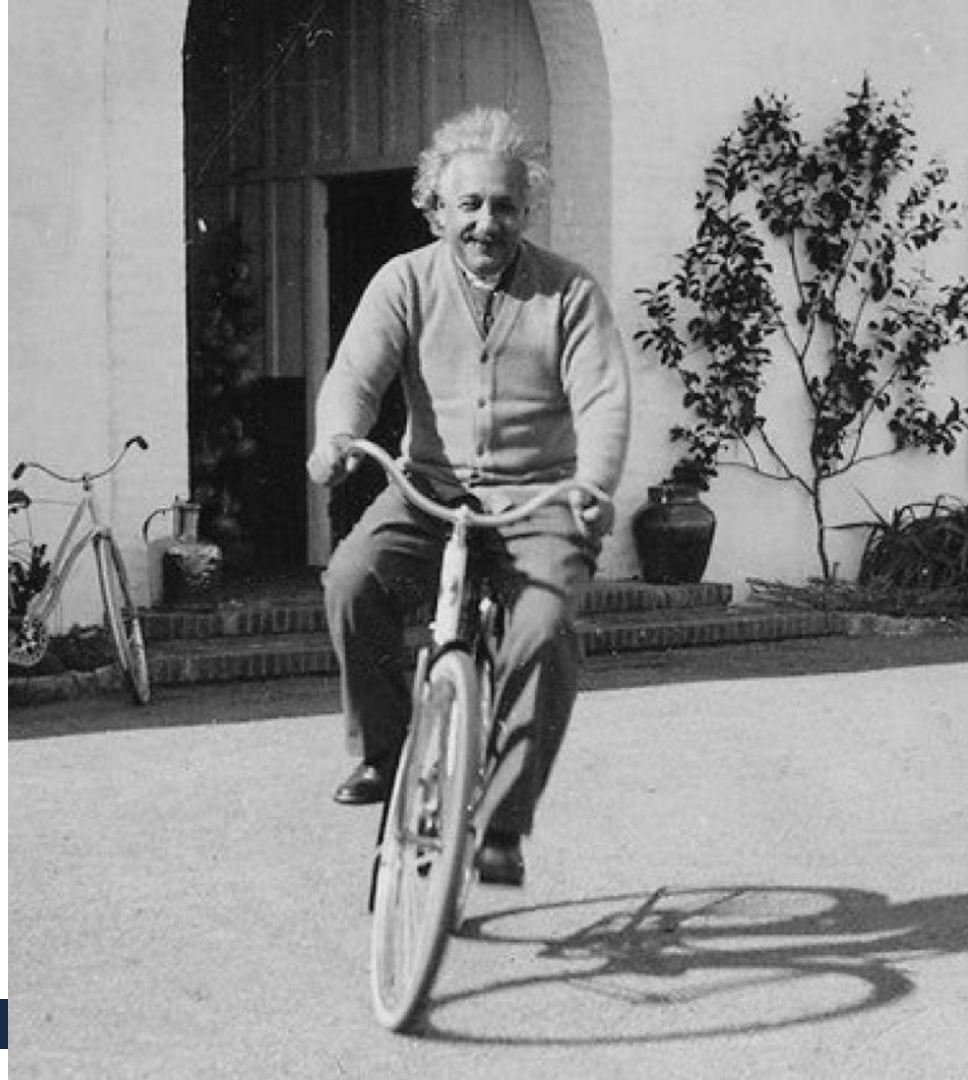
- What did we **learn**?
- Do the results make **sense**?
- Can we tell a **story**?

The process is **iterative**: what you learn at any step can send you back to refine your question.

Adapted from Joe Blitzstein & Hanspeter Pfister, Harvard CS109

Asking Good Questions is Key!

If I had an hour to solve a problem and my life depended on it, I would use the first 55 minutes determining the proper question to ask, for once I know the proper question, I could solve the problem in less than five minutes. — attributed to Einstein



Data Science questions should...

- answerable with data it is possible to collect
- unambiguous in meaning
- specifically describing exactly what data/metrics/analysis are required to answer the question
- big enough to be interesting, small enough to be accomplishable

You often may not know if it is answerable, but you get better at this. You can also elect to buy/gather more data!

Specifying what you're going to measure is important:

Examples of poor questions that leave wiggle room for useless answers:

- What can my data tell me about my business?
- How can I increase my profits?

Examples of good questions where the answer is impossible to avoid:

- How many Cybertrucks will Tesla sell in San Diego during the third quarter?
- What is the relationship between how many students are offered admissions at UCSD and how many enroll?

Working toward a strong data science question…

Nailing down the right question: **politics**

Too-vague: What impacts American politics?

… Does pop culture have an impact on American politics?

… Do American TV shows have an *impact* on American politics?

… Does South Park *affect* American politics?

… Is there a *relationship between* words in South Park episodes and American politics?

… Is there a relationship between the sentiment of political words in South Park and American politics?

Better: What is the relationship between the sentiment of political words in South Park and America's presidential approval rating?

Nailing down the right question: education

Too-vague: How has COVID-19 impacted students?

Improving: How has COVID-19 impacted university students' education?

… Do students' grades and how they rate their classes differ pre- and during remote learning, due to COVID-19?

… At UCSD, is there a difference between students' grades and how they rate their classes before COVID-19 and during remote learning, due to COVID-19?

Better: At UCSD, comparing remote learning during COVID-19 to the two years pre-COVID, what was the effect on students' grades and how students rate their courses?

https://forms.gle/UvgrcurFqcVrjcJt7

Data Science Question

Here is a too-vague question:

What racial disparities exist in policing?

Let's refine it…Which of the following would be the best possible data science question?

https://forms.gle/MjvC7uj5qMXD2EiCA

Think-Pair-Share: $an Diego Co$tS

Too vague question: Why is it so expensive to live in San Diego?

1P:

General topics:

More specific question:

2PM:

General:

More specific question:

What happens next?

After the question is defined, should it become a project?

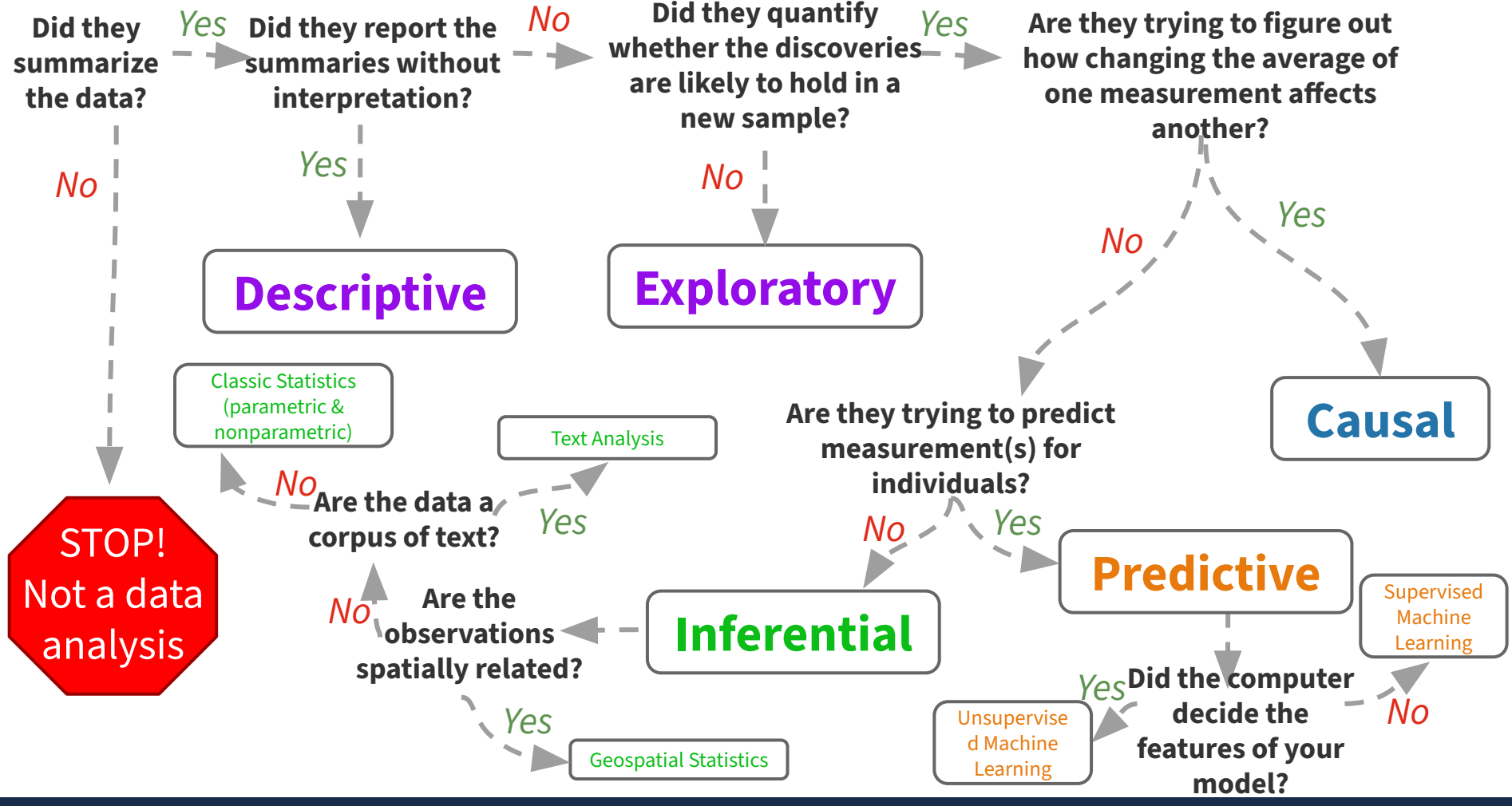
- What are the constraints?
- What are the resources available?
- ARE THE NECESSARY DATA GETTABLE? - will need this for first checkpoint
- Costs? Benefits?
- Potential risks? Rewards? (Ethical)
- Can we define a metric to determine success of the project?

"Data science is the process of formulating a quantitative question that can be answered with data, collecting and cleaning the data, analyzing the data, and communicating the answer to the question to a relevant audience."



To do this, you have to *look at, describe, and explore* the data

https://simplystatistics.org/2015/03/17/data-science-done-well-looks-easy-and-that-is-a-big-problem-for-data-scientists/


Did they summarize the data?
Yes
Did they report the summaries without interpretation?
No
Did they quantify whether the discoveries are likely to hold in a new sample?
Yes
Are they trying to figure out how changing the average of one measurement affects another?
No
Yes
Causal
No
Descriptive
Yes
No
Exploratory
Yes
Are they trying to predict measurement(s) for individuals?
No
Yes
Predictive
Supervised Machine Learning
Did the computer decide the features of your model?
Yes
Unsupervised Machine Learning
No
Inferential
Are the observations spatially related?
No
Yes
Geospatial Statistics
Are the data a corpus of text?
No
Classic Statistics (parametric & nonparametric)
Yes
Text Analysis
No
STOP! Not a data analysis

Summary: Analytical Approaches

1. Exploratory Data Analysis is the first step
2. Inference establishes relationships
   a. Classic Statistics
   b. Geospatial Analysis
   c. Text Analysis
3. Machine Learning is for prediction
   a. Supervised
   b. Unsupervised
4. Experiments best way to establish causality

Your final project can ask an inferential, predictive, (or sometimes causal) question. You and your group decide!

General question: What impacts politics in America?

Data Science question: Is there a relationship between the sentiment of political words in South Park and America's presidential approval rating?

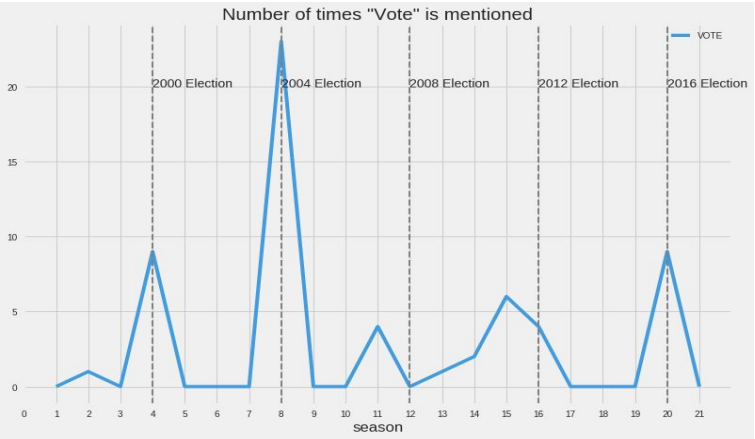
Descriptive

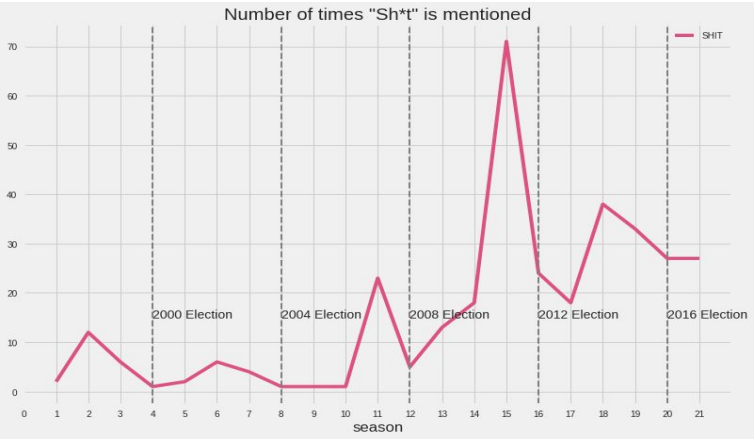
Exploratory

Inferential

Text Analysis

Classic Statistics (parametric & nonparametric)





https://github.com/COGS108/FinalProjects-Wi18/blob/master/071-FinalProject.ipynb

General question: How has COVID-19 impacted students?

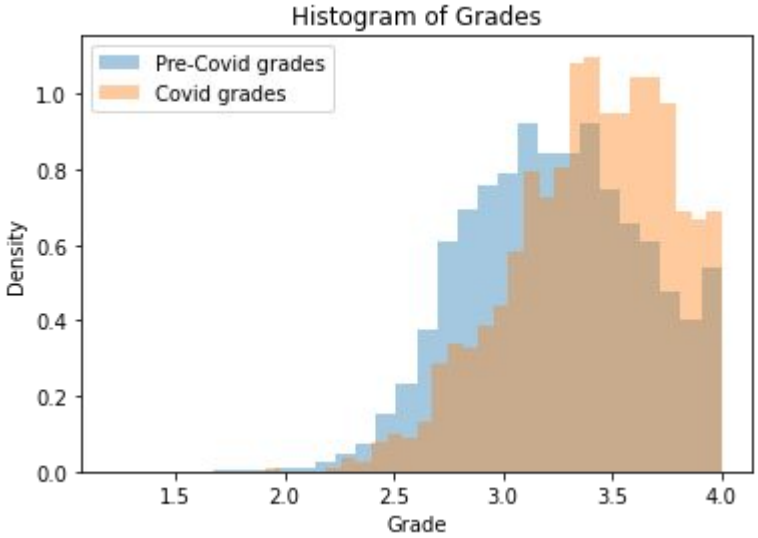
Data Science question: At UCSD, comparing remote learning during COVID-19 to the two years pre-COVID, what was the effect on students' grades and how students rate their courses?

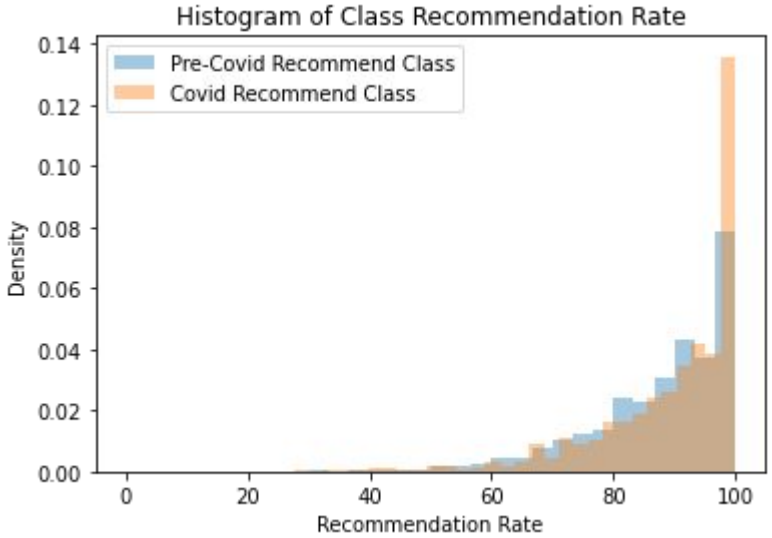
Descriptive

Exploratory

Inferential

Classic Statistics (parametric & nonparametric)

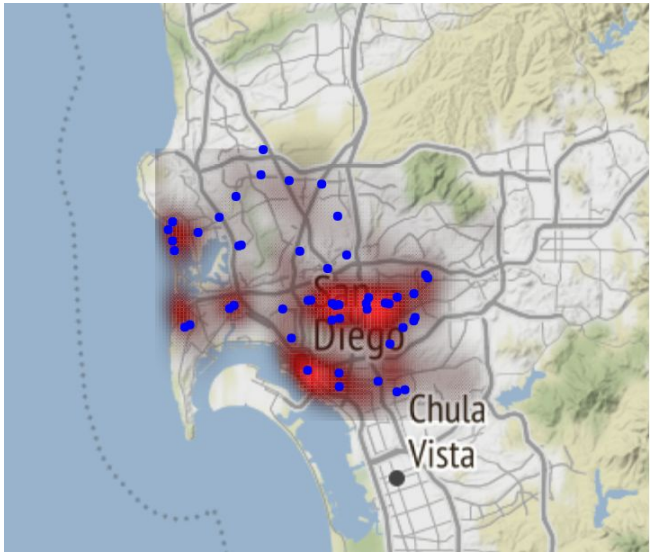


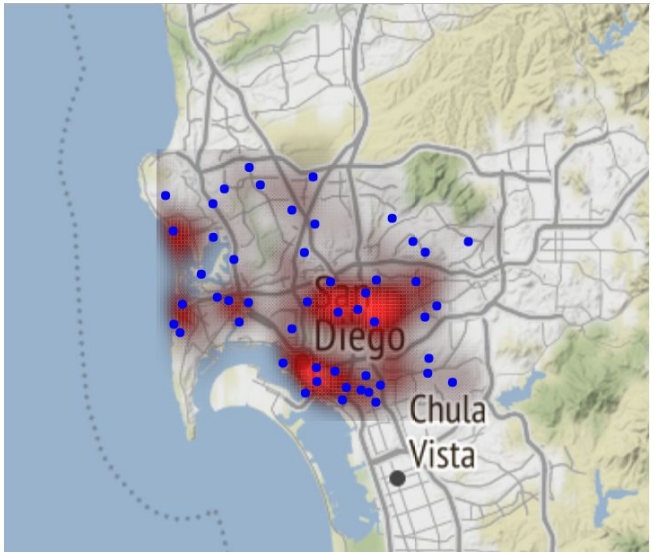


https://github.com/COGS108/FinalProjects-Wi21/blob/main/FinalProject_group084.ipynb

General question: Why isn't police response time always the same?

Data Science question: Where should police cars be stationed, accounting for crime levels and time of day, to make police response times equitable throughout San Diego?





Descriptive

Exploratory

Predictive

Inferential

Geospatial Analysis

https://github.com/COGS108/FinalProjects-Sp17/blob/master/029-FinalProject.ipynb